

SEUNGPIL LEE

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Personal Statement

Master’s researcher who moves in both directions between AI and the human mind: from AI back to the mind it imitates, and from the mind forward into the AI it shapes. My current focus is the cognition of intelligent systems, studied through LLMs — mechanistic interpretability keeps me honest about mechanisms, and memory-augmented agents that learn at test time are the current testbed. Coming in through literature and philosophy of mind, I draw on cognitive science and STS alongside the engineering.

EDUCATION

Gwangju Institute of Science and Technology (GIST) Master’s Program, Major in AI Convergence	<i>March 2025 – present</i>
Gwangju Institute of Science and Technology (GIST) B.S. in Electrical Engineering and Computer Science Minor in Mathematics, Minor in Literature	<i>March 2018 – December 2024</i> GPA: 3.66/4.50
UC Berkeley Berkeley Summer Session Program	<i>June 2019 – August 2019</i> GPA: 4.00/4.00

PUBLICATIONS

Journal Articles

- **Seungpil Lee***, Woochang Sim*, Donghyeon Shin*, Sanha Hwang, Wongyu Seo, Jiwon Park, Seokki Lee, Sejin Kim, and Sundong Kim. “Reasoning Abilities of Large Language Models: In-Depth Analysis on the Abstraction and Reasoning Corpus.” *ACM Transactions on Intelligent Systems and Technology* (**ACM TIST**), 2024.
- Sanha Hwang, **Seungpil Lee**, Sejin Kim, and Sundong Kim. “Solution Augmentation for ARC-AGI Problems Using GFlowNet: A Probabilistic Exploration Approach.” *Transactions on Machine Learning Research* (**TMLR**), 2025.

Conference Papers

- **Seungpil Lee***, Donghyeon Shin*, Klea Lena Kovacec, and Sundong Kim. “From Generation to Selection: Findings of Converting Analogical Problem-Solving into Multiple-Choice Questions.” *Findings of EMNLP* (**EMNLP Findings**), 2024.
- Hosung Lee*, Sejin Kim*, **Seungpil Lee**, Sanha Hwang, Jihwan Lee, Byung-Jun Lee, and Sundong Kim. “ARCLE: The Abstract and Reasoning Corpus Learning Environment for Reinforcement Learning.” *Conference on Lifelong Learning Agents* (**CoLLAs**), 2024.

Preprints

- Heejun Kim*, **Seungpil Lee***, Jewon Yeom, Jaewon Sok, Seonghyeon Park, Jeongjae Park, Taesup Kim, and Sundong Kim. “From Noise to Diversity: Random Embedding Injection in LLM Reasoning.” *arXiv preprint*, 2026. (**Accepted at ICML Mechanistic Interpretability Workshop 2026, Spotlight**)
- **Seungpil Lee**, Donghyeon Shin, Yunjeong Lee, and Sundong Kim. “Can Large Language Models Develop Gambling Addiction?” *arXiv preprint*, 2025.

Workshop Papers

- **Seungpil Lee***, Woochang Sim*, Donghyeon Shin*, Sejin Kim, and Sundong Kim. “Reasoning Abilities of Large Language Models through the Lens of Abstraction and Reasoning.” *NeurIPS Workshop on System-2 Reasoning at Scale*, 2024.
- Donghyeon Shin*, **Seungpil Lee***, Klea Lena Kovacec, and Sundong Kim. “Regulation Using Large Language Models to Generate Synthetic Data for Evaluating Analogical Ability.” *IJCAI Workshop on Analogical Abstraction*, 2024.

Domestic Publications

- **Seungpil Lee**, Jihwan Lee, and Sundong Kim. “Evaluating Prior Knowledge of ARC Using World Models.” *Korea Software Congress*, 2023.
- Jihwan Lee, **Seungpil Lee**, Sejin Kim, and Sundong Kim. “Extracting the Core Knowledge of ARC with the World Model.” *Korea Software Congress*, 2023.
- Donghyeon Shin, Sanha Hwang, Seokki Lee, Yunho Kim, **Seungpil Lee**, and Sundong Kim. “MC-LARC Benchmark to Measure LLM Reasoning Capability.” *Korea Software Congress*, 2023.

* indicates equal contribution

SKILLS

Programming	C, C++, Java, JavaScript, Python
Tools	LaTeX, Spring Boot, PyTorch, Hugging Face
Language	Korean (Native), English (Intermediate)

EXPERIENCE

LG AI Research, Seoul <i>Research Intern</i>	<i>September 2026 – present</i>
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- Improving the coding capability of K-Exaone, a Korean sovereign foundation model

Microsoft Research Asia (MSRA), Shanghai <i>Research Intern, Collaborative Research Program</i>	<i>February 2026 – August 2026</i>
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- Developed test-time skill optimization for LLM agents, letting them acquire and reuse task skills at inference time without parameter retraining
- Extended the Agent-Lightning framework with memory generation so agents carry learned context across long-horizon tasks
- Bridged mechanistic analysis of reasoning failures with memory-based mitigations

DataScience Lab, GIST <i>Undergraduate Intern → Master’s Student</i>	<i>September 2023 – present</i>
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- LLM skill discovery and evaluation with cognitive science tools (Language of Thought Hypothesis)
- LLM alignment using RLVR methods; Mechanistic interpretability with Sparse Autoencoders

Teaching Assistant, Computer Networking	<i>September 2023 – December 2025</i>
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- Designed assignments connecting theory to implementation; SDN demo with Ryu and Mininet

Development Team, Republic of Korea Navy	<i>January 2022 – September 2023</i>
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- Mandatory military service; back-end developer for internal app and web services

BioComputing Lab, GIST <i>Undergraduate Intern</i>	<i>June 2020 – January 2021</i>
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- Developed Spiking Neural Network methods with synthetic gradients

AWARDS & FUNDING

National Research Foundation of Korea (NRF) Funding Research funding for outstanding master's students	<i>Sep 2025 – Sep 2026</i>
Best Undergraduate Thesis Award, GIST EECS Awarded for graduation thesis on MC-LARC benchmark	<i>February 2025</i>
Korean Government Scholarship, GIST College Full scholarship for students at GIST	<i>March 2018 – August 2024</i>
Scholarship for Summer Session Abroad UC Berkeley Summer Session scholarship	<i>June – August 2019</i>
Navy AI Competition Award in drone object detection competition	<i>August 2023</i>
K-StartUp Finals Advanced to finals as ML researcher and back-end developer	<i>November 2023</i>

EXTRA-CURRICULAR

Published Author – Short Story Collection Published a 400-page short story collection exploring consciousness and identity	<i>2019</i>
AGIST – Deep Learning Study Group Presentations on Spiking Neural Networks with STDP learning	<i>August 2020 – January 2021</i>